

Exercise

Outlook	Temp	Humidity	Windy	Class
sunny	75	70	TRUE	Play
sunny	80	90	TRUE	Don't Play
sunny	85	85	FALSE	Don't Play
sunny	72	95	FALSE	Don't Play
sunny	69	70	FALSE	Play
?	72	90	TRUE	Play
overcast	83	78	FALSE	Play
overcast	64	65	TRUE	Play
overcast	81	75	FALSE	Play
rain	71	80	TRUE	Don't Play
rain	65	70	TRUE	Don't Play
rain	75	80	FALSE	Play
rain	68	80	FALSE	Play
rain	70	96	FALSE	Play

1. What is GainRatio for Outlook?
2. What are Examples (instances) for Outlook = sunny? (if "Outlook" is the root)
3. Based on the result of question (2), if the next attribute is "Humidity" with threshold ≤ 75 and >75 , illustrate the Decision Tree for that branch (assumption: stop learning after "Humidity" is selected)
4. If we have unseen data: $\langle \text{Outlook} = \text{sunny}, \text{Temp} = 70, \text{Humidity} = ?, \text{and Windy} = \text{FALSE} \rangle$, what is the class prediction? (Based on the result of question (3)).

Exercise

4

Outlook	Temp	Humidity	Windy	Class
sunny	75	70	TRUE	Play
sunny	80	90	TRUE	Don't Play
sunny	85	85	FALSE	Don't Play
sunny	72	95	FALSE	Don't Play
sunny	69	70	FALSE	Play
?	72	90	TRUE	Play
overcast	83	78	FALSE	Play
overcast	64	65	TRUE	Play
overcast	81	75	FALSE	Play
rain	71	80	TRUE	Don't Play
rain	65	70	TRUE	Don't Play
rain	75	80	FALSE	Play
rain	68	80	FALSE	Play
rain	79	96	FALSE	Play

Continuous-valued attribute

Handling missing attribute value

Handling attributes with differing costs

1. What is **GainRatio** for **Outlook**?
2. What are Examples (instances) for Outlook = sunny?
3. Based on the result of question (2), define the threshold in "Humidity" discretization, and the leaf node for each branch.

Gain(S,A) only consider the fraction of training examples with known value

$$\text{Gain}(S,A) = |I|/|I| * (\text{Entropy}(S) - [\sum \text{proportion} * \text{entropy_of_known_value}])$$

$$\text{GainRatio}(S, A) \equiv \frac{\text{Gain}(S, A)}{\text{SplitInformation}(S, A)}$$

$$\text{SplitInformation}(S, A) \equiv - \sum_{i=1}^c \frac{|S_i|}{|S|} \log_2 \frac{|S_i|}{|S|}$$

set of value v_i

Tanpa missing values

$$\text{Entropy}(S) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.940$$

$$\begin{aligned} 5 \quad \text{Gain}(S, \text{Outlook}) &= \text{Entropy}(S) - \sum_{\text{value of Outlook}} \frac{S_v}{S} \text{Entropy}(S_v) \\ &= 0.940 - \left[\frac{5}{14} \text{Entropy}(\text{Sunny}) + \frac{4}{14} \text{Entropy}(\text{Overcast}) + \frac{5}{14} \text{Entropy}(\text{Rain}) \right] \\ &= 0.940 - \left[\left(\frac{5}{14} \left(-\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} \right) \right) + \left(\frac{4}{14} \left(-\frac{4}{4} \log_2 \frac{4}{4} - \frac{0}{4} \log_2 \frac{0}{4} \right) \right) + \left(\frac{5}{14} \left(-\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} \right) \right) \right] = 0.246 \end{aligned}$$

Handling missing attribute value

$$\text{Gain}(S, A) = 13/14 * (\text{Entropy}(S) - [\sum \text{proportion} * \text{entropy_of_known_value}])$$

$$\text{Entropy}(S) = -\frac{8}{13} \log_2 \frac{8}{13} - \frac{5}{13} \log_2 \frac{5}{13} = 0.961$$

$$\begin{aligned} \text{Gain}(S, \text{Outlook}) &= 13/14 * (0.961 - [\frac{5}{13} \text{Entropy}(\text{Sunny}) + \frac{3}{13} \text{Entropy}(\text{Overcast}) + \frac{5}{13} \text{Entropy}(\text{Rain})]) \\ &= 13/14 * (0.961 - [\frac{5}{13} (-\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5}) + \frac{3}{13} (-\frac{3}{3} \log_2 \frac{3}{3} - \frac{0}{3} \log_2 \frac{0}{3}) + \frac{5}{13} (-\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5})]) \end{aligned}$$

$$= 13/14 * (0.961 - [\frac{5}{13} 0.971 + \frac{3}{13} 0 + \frac{5}{13} 0.971]) = 0.199$$

$$\text{SplitInformation}(S, \text{Outlook}) = -\sum_{i=1}^4 \frac{|S_i|}{|S|} \log_2 \frac{|S_i|}{|S|} = -(\frac{5}{14} \log_2 \frac{5}{14} + \frac{3}{14} \log_2 \frac{3}{14} + \frac{5}{14} \log_2 \frac{5}{14} + \frac{1}{14} \log_2 \frac{1}{14}) = 1.809$$

$$\text{GainRatio}(S, \text{Outlook}) = \frac{\text{Gain}(S, \text{Outlook})}{\text{SplitInformation}(S, \text{Outlook})} = \frac{0.199}{1.809} = 0.110$$

Exercise

6

Outlook	Temp	Humidity	Windy	Class
sunny	75	70	TRUE	Play
sunny	80	90	TRUE	Don't Play
sunny	85	85	FALSE	Don't Play
sunny	72	95	FALSE	Don't Play
sunny	69	70	FALSE	Play
?	72	90	TRUE	Play
overcast	83	78	FALSE	Play
overcast	64	65	TRUE	Play
overcast	81	75	FALSE	Play
rain	71	80	TRUE	Don't Play
rain	65	70	TRUE	Don't Play
rain	75	80	FALSE	Play
rain	68	80	FALSE	Play
rain	70	86	FALSE	Play

Termasuk isinya?

Alternative measures for selecting attributes

Overfitting training data

Continuous-valued attribute

Handling missing attribute values

Handling attributes with differing costs

1. What is GainRatio for Outlook?
2. What are Examples (instances) for Outlook = sunny?
3. Based on the result of question (2), define the threshold in "Humidity" discretization, and the leaf node for each branch.

Outlook	Temp	Humidity	Windy	Class	Weight
sunny	75	70	TRUE	Play	1
sunny	80	90	TRUE	Don't Play	1
sunny	85	85	FALSE	Don't Play	1
sunny	72	95	FALSE	Don't Play	1
sunny	69	70	FALSE	Play	1
?	72	90	TRUE	Play	5/13

↳ missing value

Selalu masuk,
kalaupun dia tanya outlook: overcast, di
missing value jg masuk tp bobotnya
berbeda.

Exercise

7

Outlook	Temp	Humidity	Windy	Class	Weight
sunny	75	70	TRUE	Play	1
sunny	80	90	TRUE	Don't Play	1
sunny	85	85	FALS E	Don't Play	1
sunny	72	95	FALS E	Don't Play	1
Humidity		Class			
70		Play	70	E	Play
70		Play	90	TRUE	Play
Don't					
85		Play			
Don't					
90		Play			
90		Play			
Don't					
95		Play			

1. What is GainRatio for Outlook?
2. What are Examples (instances) for Outlook = sunny?
3. Based on the result of question (2), define the threshold in "Humidity" discretization, and the leaf node for each branch.
 $\Rightarrow$ Threshold = $(70+85)/2=77.5$

...

Outlook=sunny:

- humidity < 77.5 = True: Play
- humidity < 77.5 = False: <5/13 Play, 3 Don't Play>

LATIHAN MINGGU 4

Jika terdapat training examples sebagai berikut, pembatas (threshold) yang mungkin diperiksa untuk attr2 adalah:

sortir nilai attr2 dulu

Instance	attr1	attr2	attr3	Label
1	v1_1	8	v3_1	0
2	v1_2	5	v3_1	1
3	v1_1	5	v3_2	1
4	v1_2	7	v3_1	1
5	v1_3	6	v3_1	0
6	v1_3	4	v3_2	0
7	v1_3	5	v3_1	1
8	v1_1	8	v3_1	0
9	v1_2	5	v3_2	1
10	v1_3	6	v3_1	0

- 1 pembatas yaitu ($\text{attr2} \leq 6$ dan $\text{attr2} > 6$)
- 2 pembatas yaitu ($\text{attr2} \leq 5$ dan $\text{attr2} > 5$), ($\text{attr2} \leq 7$ dan $\text{attr2} > 7$)
- 3 pembatas yaitu ($\text{attr2} \leq 5$ dan $\text{attr2} > 5$), ($\text{attr2} \leq 6$ dan $\text{attr2} > 6$), ($\text{attr2} \leq 7$ dan $\text{attr2} > 7$)
- 4 pembatas yaitu ($\text{attr2} \leq 4$ dan $\text{attr2} > 4$), ($\text{attr2} \leq 5$ dan $\text{attr2} > 5$), ($\text{attr2} \leq 6$ dan $\text{attr2} > 6$), ($\text{attr2} \leq 7$ dan $\text{attr2} > 7$)

semua pilihan a, b, c, d ini salah,
cuman yg paling mendekati = d. cuman
batasnya harusnya 4.5, 5.5, 6.5, dll
(lihat slide bawah)

LATIHAN MINGGU 4

2. Jika terdapat training examples sebagai berikut, pembatas (threshold) yang mungkin diperiksa untuk attr2 adalah:

Instance	attr1	attr2	attr3	Label
1	v1_1	8	v3_1	0
2	v1_2	5	v3_1	1
3	v1_1	5	v3_2	1
4	v1_2	7	v3_1	1
5	v1_3	6	v3_1	0
6	v1_3	4	v3_2	0
7	v1_3	5	v3_1	1
8	v1_1	8	v3_1	0
9	v1_2	5	v3_2	1
10	v1_3	6	v3_1	0

Attr2	Label
4	0
5	1
5	1
5	1
5	1
6	0
6	0
7	1
8	0
8	0

$$\frac{4+5}{2} = 4.5$$

Kandidat 1:
4.5

Kandidat 2:
5.5

Kandidat 3:
6.5

Kandidat 4:
7.5